TS8 - TC2 Lucas Gallero

(1)

Ceros $\omega_0 = 1 \text{ rad/seg}$ $\alpha = 0,5 \text{ rad/seg}$

$$S_{z_1,z_2} = -\alpha \pm j\sqrt{\omega_0^2 - \alpha^2} = \boxed{-0,5 \pm j\,0,866}$$

Polos $\boxed{S_{p_1} = -1}$ $S_{p_{2,3}} = 5e^{j116,6°} = \overbrace{5\cos(116,6°)}^{\alpha} + j\overbrace{5\sin(116,6°)}^{\omega}$

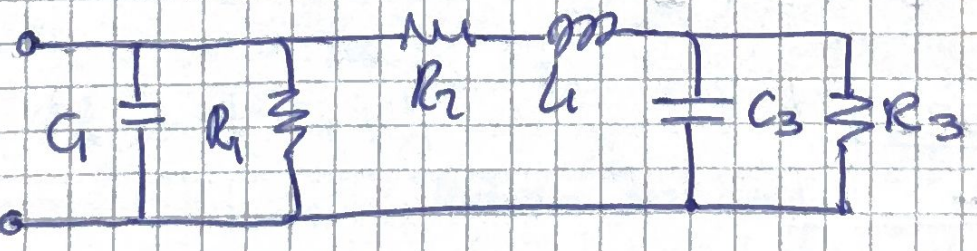
$$\boxed{S_{p_{2,3}} \cong -2,239 \pm j\,4,471}$$

▶ $-\alpha \pm j\omega \rightarrow s^2 + 2\alpha s + (\alpha^2 + \omega^2)$

Ceros $\rightarrow s^2 + s + 1$

Polos $\rightarrow (s+1)(s^2 + 4,4776s + 25)$

$$\boxed{Z(s) = k\,\frac{(s^2+s+1)}{(s+1)(s^2+4,4776s+25)}}$$



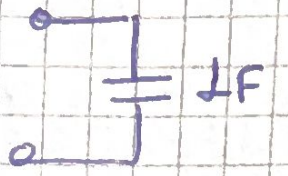
$\boxed{k = 1}$

$$\boxed{Z(s) = \frac{s^2+s+1}{s^3+5,4775s^2+29,4775s+25}}$$

NOTA

$$Y(s) = \frac{s^3 + 5{,}477s^2 + 29{,}477s + 25}{s^2 + s + 1}$$

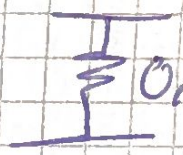
$$C_1 = \lim_{s\to\infty} \frac{1}{s} Y(s) = 1F$$



$$Y_2 = \frac{\cancel{s^3} + 5{,}477s^2 + 29{,}477s + 25 - \cancel{s^3} - s^2 - s}{s^2 + s + 1}$$

$$Y_2 = \frac{4{,}477s^2 + 28{,}477s + 25}{s^2 + s + 1}$$

$$\frac{1}{R_1} = \lim_{s\to\infty} Y_2(s) = 4{,}477 \longrightarrow R_1 = 0{,}223$$



$$Y_4(s) = Y_2(s) - \frac{1}{R_1}$$

$$Y_4(s) = \frac{4{,}477s^2 + 28{,}477s + 25}{s^2 + s + 1} - 4{,}477$$

$$Y_4(s) = \frac{\cancel{4{,}477s^2} + 28{,}477s + 25 - (\cancel{4{,}477s^2} + 4{,}477s + 4{,}477)}{s^2 + s + 1}$$

$$Y_4(s) = \frac{24s + 20{,}522}{s^2 + s + 1}$$

$$Z_4(s) = \frac{s^2 + s + 1}{24s + 20{,}522}$$

$$L_1 = \lim_{s\to\infty} Z_4(s) \cdot \frac{1}{s} = \frac{s^2 + s + 1}{24s^2 + 20{,}522s}\Big|_{\infty} = \frac{1}{24} = 0{,}04166.$$

NOTA

HOJA N°

FECHA

$$Z_6 = Z_4 - S\cdot\frac{1}{24} = \frac{S^2+S+1 - S\frac{1}{24}(24S+20{,}522)}{24S+20{,}522}$$

$$Z_6 = \frac{0{,}1449\,S+1}{24(S+0{,}855)}$$

$$R_2 = \lim_{S\to\infty} Z_3(S) = \frac{0{,}1449}{24} = 0{,}006$$

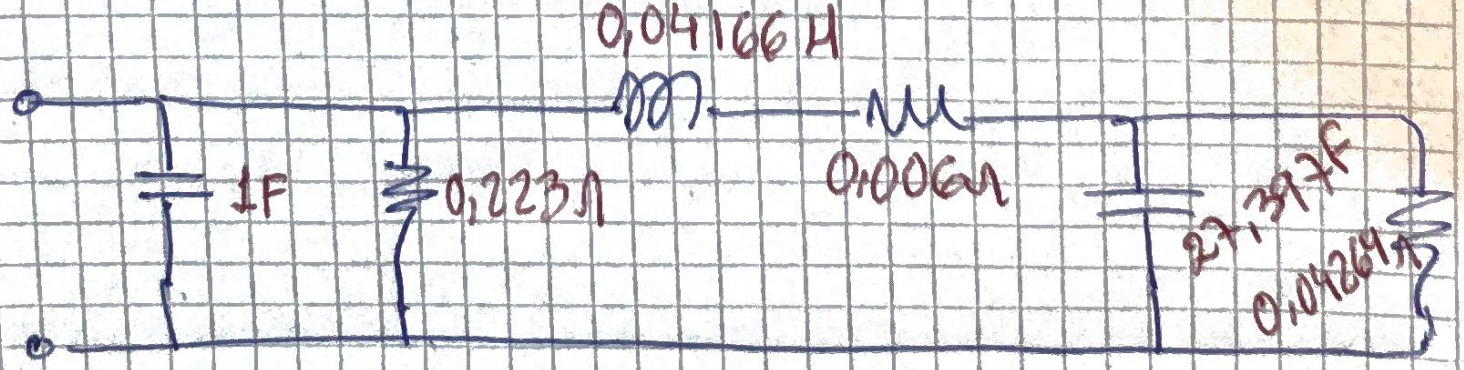
$$Z_8 = \frac{0{,}1449\,S+1}{24(S+0{,}855)} - \frac{0{,}1449}{24} = \frac{0{,}1449\,S+1-0{,}1449(S+0{,}855)}{24(S+0{,}855)}$$

$$Z_8 = \frac{0{,}0365}{S+0{,}855} = \frac{1}{SC+\frac{1}{R}} = \frac{\frac{1}{C}}{S+\frac{1}{RC}}$$

$$C_3 = \frac{1}{0{,}0365} = 27{,}397$$

$$R_3 = \frac{1}{(0{,}855)(27{,}397)} = 0{,}04269$$

Resumen

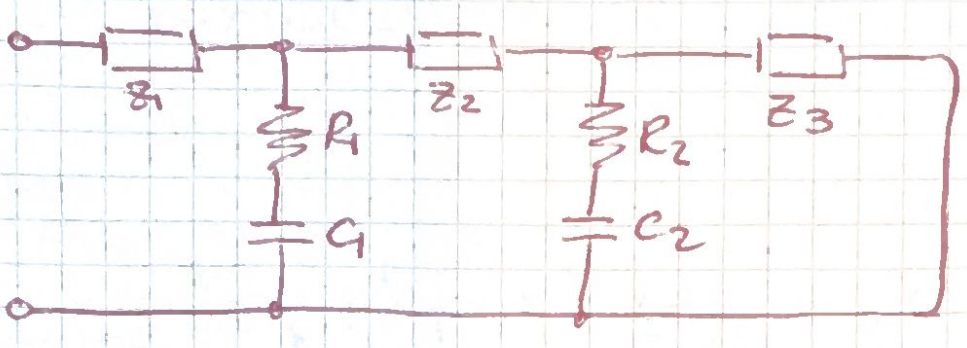


NOTA

TS8 - TC2

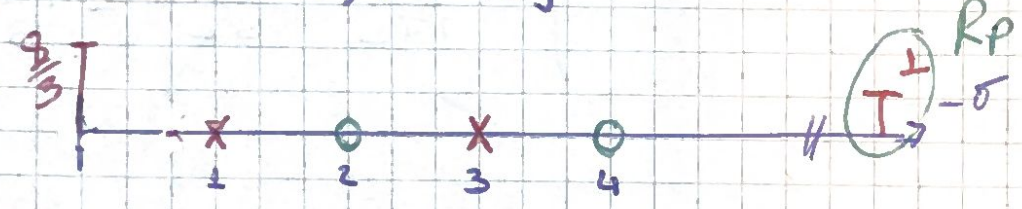
(2)

$C_1R_1 = 1/5 \qquad C_2R_2 = 1/4 \qquad Z(S) = \dfrac{S^2+6S+8}{S^2+4S+3}$



$Z(S) = \dfrac{(S+2)(S+4)}{(S+1)(S+3)}$

o En Y Tengo que sacar en (S+5)

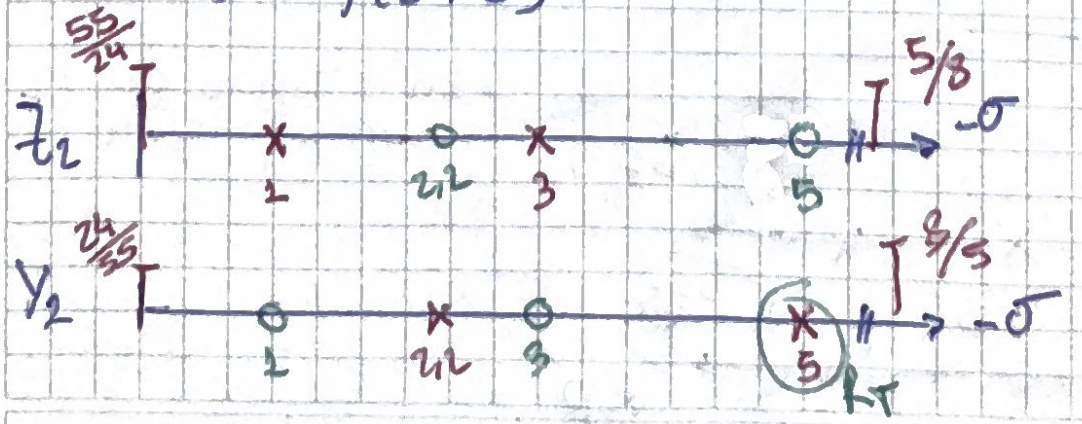


$K_\infty = \lim_{S\to -5} Z(S) = \lim_{S\to -5} \dfrac{(-5)^2+6(-5)+8}{(-5)^2+4(-5)+3} = \dfrac{3}{8} = 0{,}375$

$Z_2 = Z - Z_1 = \dfrac{S^2+6S+8}{S^2+4S+3} - 3/8 = \dfrac{S^2+6S+8-3/8\,S^2+3/2\,S+9/8}{S^2+4S+3}$

$Z_2 = \dfrac{5/8\,S^2+9/2\,S+55/8}{S^2+4S+3} = \dfrac{5/8\,(S^2+36/5\,S+11)}{(S^2+4S+3)}$

$Z_2 = \dfrac{5/8\,(S+11/5)(S+5)}{(S+1)(S+3)}$



NOTA

$$Y_2(s) = \frac{8}{5} \cdot \frac{(s+1)(s+3)}{(s+4/5)(s+5)}$$

$$k_1 = \lim_{s \to -5} \cdot \frac{s+5}{s} \cdot \frac{8}{5} \frac{(s+1)(s+3)}{(s+4/5)(s+5)} = \frac{8}{5} \frac{(-5+1)(-5+3)}{(-5)(-5+4/5)}$$

$$\boxed{k_1 = \frac{32}{35}}$$

$$\text{Lo} \quad Y_{RO} = \frac{s k_1}{s+\sigma_1} = \frac{1}{\frac{1}{k_1} + \frac{1}{s\, k_1/\sigma_1}}$$

$$C = \frac{k}{\sigma} = \frac{32/35}{5} = \frac{32}{175}$$

$$R = \frac{1}{k} = \frac{35}{32}.$$

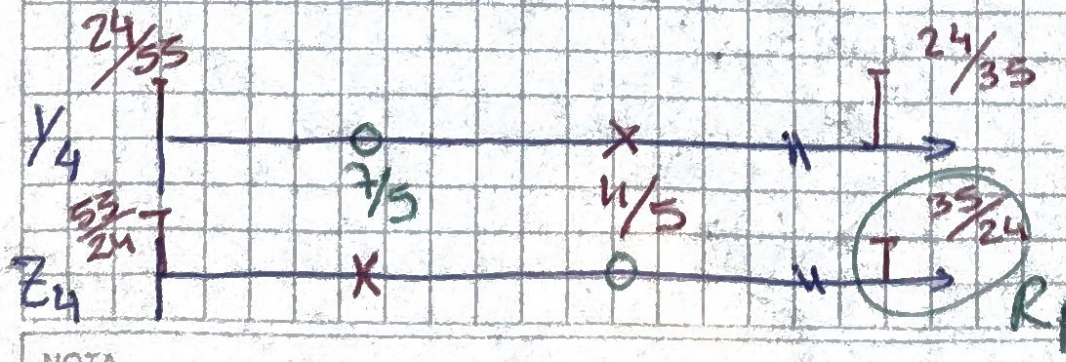
$$\frac{32}{175} \cdot \frac{35}{32} = \frac{1}{5}$$ ✓

$$Y_4 = Y_2 - X_3 = \frac{s^2 + 4s + 3}{5/8\,(s+4/5)(s+5)} - \frac{\frac{32}{35} s}{(s+5)}$$

$$Y_4 = \frac{s^2 + 4s + 3 - \frac{32}{35} s \cdot 5/8 (s+4/5)}{5/8\,(s+4/5)(s+5)} = \frac{\frac{3}{7} s^2 + \frac{96}{35} + 3}{5/8\,(s+4/5)(s+5)}$$

$$Y_4 = \frac{\frac{3}{7}\left(s^2 + \frac{32}{5} s + 7\right)}{5/8\,(s+4/5)(s+5)} = \frac{24}{35} \frac{(s+7/5)(s+5)}{(s+4/5)(s+5)}$$

$$Y_4 = \frac{24}{35} \frac{(s+7/5)}{(s+4/5)}$$



• En Y tengo que sacar en $s \neq 4$

NOTA

$$Z_4 = \frac{35}{24}\,\frac{(s+11/5)}{(s+7/5)}$$

$$K_{\infty,2} = \lim_{s\to -4} \frac{35}{24}\,\frac{(s+11/5)}{(s+7/5)} = \frac{35}{24}\,\frac{(-4+11/5)}{(-4+7/5)} = \frac{105}{104}$$

$$Z_6 = Z_4 - K_{\infty,2}$$

$$Z_6 = \frac{35}{24}\,\frac{(s+11/5)}{(s+7/5)} - \frac{105}{104} = \frac{s+11/5 - \frac{105}{104}\,\frac{24}{35}(s+7/5)}{\frac{24}{35}(s+7/5)}$$

$$Z_6 = \frac{\frac{4}{13}s + \frac{16}{13}}{\frac{24}{35}(s+7/5)} = \frac{\frac{4}{13}(s+4)}{\frac{24}{35}(s+7/5)} = \frac{35}{78}\,\frac{(s+4)}{(s+7/5)}$$

$$Y_6 = \frac{78}{35}\cdot\frac{(s+7/5)}{(s+4)}$$

$$K_2 = \lim_{s\to -4} \frac{(s+4)}{s}\cdot\frac{78}{35}\cdot\frac{(s+7/5)}{(s+4)} = \frac{78}{35}\,\frac{(-4+7/5)}{(-4)}$$

$$K_2 = \frac{507}{350} \longrightarrow \frac{1}{\frac{1}{K_2} + \frac{1}{s\frac{K_2}{\sigma_2}}}$$

$$\begin{cases} C_2 = \frac{K_2}{\sigma_2} = \frac{\frac{507}{350}}{4} = \frac{507}{1400} \\ R_2 = \frac{1}{K_2} = \frac{350}{507} \end{cases}$$

$$Y_8 = Y_6 - Y_4 = \frac{78}{35}\,\frac{(s+7/5)}{(s+4)} - \frac{s\,\frac{507}{350}}{(s+4)}$$

$$Y_8 = \frac{s+7/5 - \frac{35}{78}\cdot\frac{507}{350}s}{\frac{35}{78}(s+4)} = \frac{\frac{7}{20}s + 7/5}{\frac{35}{78}(s+4)}$$

$$Y_8 = \frac{\frac{7}{20}(s+4)}{\frac{35}{78}(s+4)} \longrightarrow \boxed{Y_8 = \frac{39}{50}}$$

NOTA

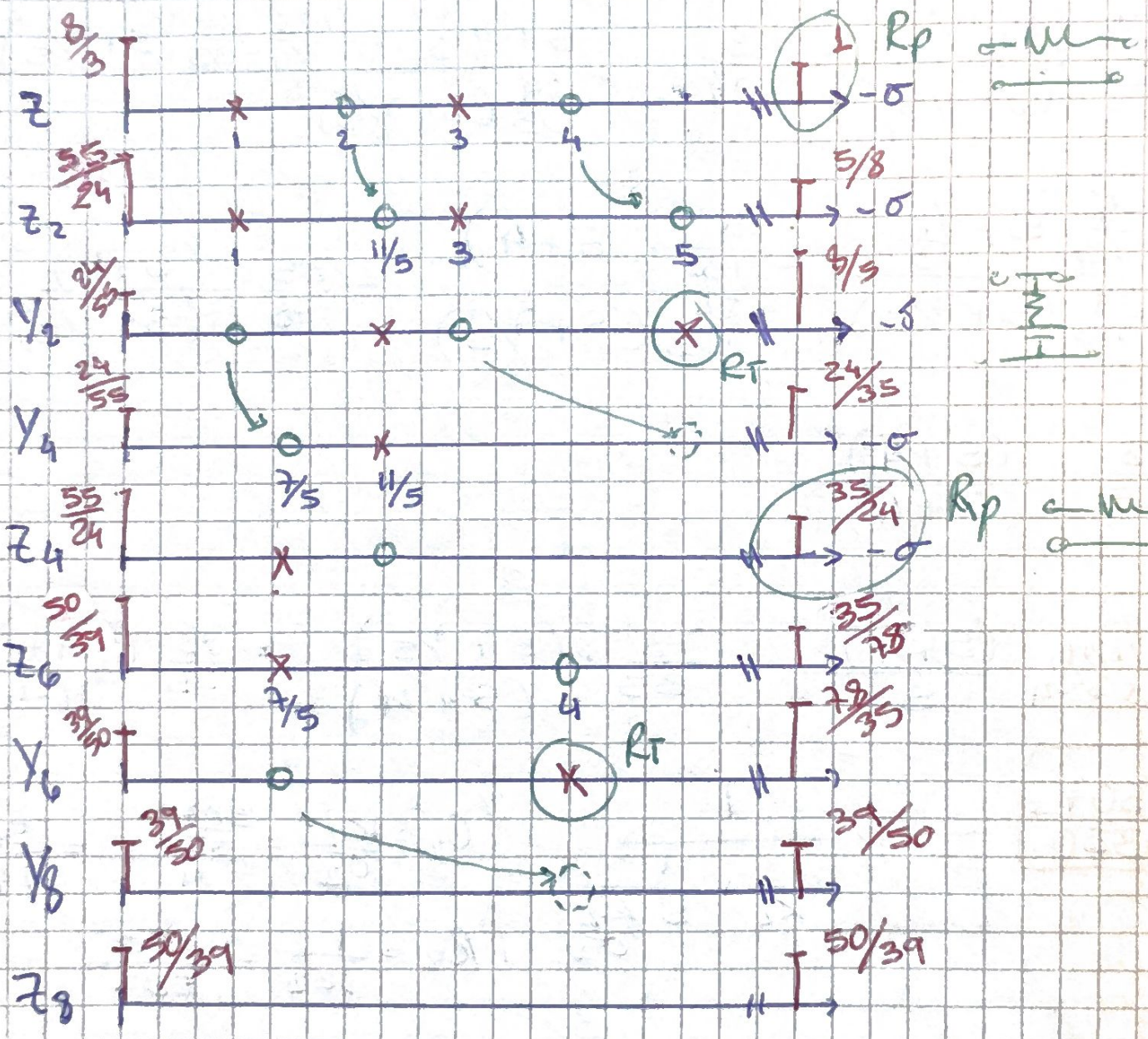
$$Z_8 = \frac{50}{39} \longrightarrow Z_3$$

Resumen

$Z_1 = 3/8$ $\quad R_1 = 35/32$ $\quad C_1 = 32/175$

$Z_2 = 105/104$ $\quad R_2 = 350/507$ $\quad C_2 = \frac{507}{1400}$

$Z_3 = 50/39$



NOTA